Pextra OS Image Compatibility Test Report

This document outlines the process and results of a compatibility test performed on the Pextra OS image within a virtualized hypervisor environment.

Objective: Validate whether the OS image functions correctly when deployed on the main hypervisor.

Compatibility Checks: Verifying boot stability, network configuration, service initialization, and overall system responsiveness. Any issues, warnings, or crashes encountered during the process were reviewed through system logs and documented below where relevant.

Reason: The results of this assessment serve as a proof-of-concept for successful deployment and provide a baseline for determining whether the Pextra platform is suitable for further testing or production-level integration within the virtualized infrastructure.

1. Environment

Hypervisor: KVM (Linux virtualization)

Operating System: Debian 12 Bookworm (64-bit)

VM Configuration: Bridged networking on LAN (*192.168.50.0/24*)

Storage: Virtual disk (standard VM disk image)

- **SATA Port 0:** Pextra.vdi (Normal, 20.00 GB)

Installation Type: Pextra CE OSE ISO (lab deployment)

- **ISO:** pextra-ce-ose-1.0-amd64.iso
- **SHA256:** `965751a3c9d41209159546b2f46ff1386f7f4af0e44a40c16778c92671d65652 pextra-ce-ose-1.0-amd64.iso`
- **ISO.ASC:**


```
-----BEGIN PGP SIGNATURE-----

iHQEABYKAB0WIQT2yCSpW1EPSe1LDWQLT5BXx9vcQQUCaZTMWAAKCRALT5BXx9vc
QREmAPUeQxAMuQ/oEtjvHYS+uME1hxM1Mv8VVSKpcTBczJWNAQDPrPEXE7uYJ5Tn
Ijc6lQtHmZ5cFDQJ9qY1TzOCiQN4DA==
=bUi8
-----END PGP SIGNATURE-----
```


- **SHA256.ASC:**


```
-----BEGIN PGP SIGNATURE-----

iHUEABYKAB0WIQT2yCSpW1EPSe1LDWQLT5BXx9vcQQUCaZTMWAAKCRALT5BXx9vc
QVABAQCJhqDaLtdkRJN55naQtOWNK3F1KNw9rXWV9gdY4yM6+gEAqQ0/swWYE/lx
```

```
1eOjbvROahaTj9xzy9GJYqm3V/sOOQc=
=KxNP
-----END PGP SIGNATURE-----
```


Graphics Controller: VMSVGA

Network: Bridged Adapter, eno2

- **Adapter 1:** Intel PRO/1000 MT Desktop

2. Installation & Configuration

Organization Name: Lab

Organization Description: This is the default organization description created during the installation process for the lab.

Country & TimeZone: United States, Etc/Utc (UTC+0000)

License Key: 88124-*****-*****-*****-*****

Username: pceadmin

Management Interface: enp0s3

Management IP: 192.168.50.251/24

Gateway: 192.168.50.1

DNS: 1.1.1.1 (*Cloudflare*)

Host: lab.local

Notes/Observations

1. OS image installed and booted successfully

2. Initial configuration completed (IP, DNS, gateway, organization, admin user)

3. Pextra dashboard is fully accessible and operational

4. No application-level crashes observed

3. Functional Testing

Network connectivity: OK (*VM reachable on LAN IP from different device*)

Web dashboard: OK (*UI loads and responds correctly, check image*)

Basic system services: check screnshot and section 5.

Running normally for approximately 1 hour with no spike in resource utilization or connection.

Login/authentication: Successful

- **Note:** Might want to check if a paid version is needed for dynamic ABAC features.
- **Note:** Depending on how IAM admin roles/access will be bootstrapped, compatibility with a vault or third-party secrets manager may need to be checked.

4. System Logs Review

Command: `sudo systemctl status pce-boot-start-instances.service`

```
● pce-boot-start-instances.service - Start Pextra CloudEnvironment(R)
instances on boot
     Loaded: loaded (/lib/systemd/system/pce-boot-start-instances.service;
enabled; preset: enabled)
     Active: active (exited) since Wed 2026-05-13 21:47:00 UTC; 1h 6min ago
    Process: 824 ExecStartPre=/bin/sleep 10 (code=exited, status=0/SUCCESS)
    Process: 1118 ExecStart=/usr/bin/pcedaemon start-all-instances
(code=exited, status=0/SUCCESS)
   Main PID: 1118 (code=exited, status=0/SUCCESS)
        CPU: 938ms

May 13 21:47:00 lab pcedaemon[1118]: 2026-05-13T21:47:00.339Z
info[db:HealthCheck]:    Testing database connection...  +1ms
May 13 21:47:00 lab pcedaemon[1118]: 2026-05-13T21:47:00.379Z
info[db:HealthCheck]:    Database connection successful.  +35ms
May 13 21:47:00 lab pcedaemon[1118]: 2026-05-13T21:47:00.395Z
info[db:Migrator]:    Running migrations...  +1ms
May 13 21:47:00 lab pcedaemon[1118]: {
May 13 21:47:00 lab pcedaemon[1118]:   results: [],
May 13 21:47:00 lab pcedaemon[1118]: }
May 13 21:47:00 lab pcedaemon[1118]: 2026-05-13T21:47:00.502Z
info[app:CheckConfig]:    Configuration check complete.  +0ms
May 13 21:47:00 lab pcedaemon[1118]: 2026-05-13T21:47:00.503Z info:
Starting all instances with autostart=true  +1ms
May 13 21:47:00 lab pcedaemon[1118]: 2026-05-13T21:47:00.507Z info:    No
instances to start, returning  +1ms
May 13 21:47:00 lab systemd[1]: Finished pce-boot-start-instances.service -
Start Pextra CloudEnvironment(R) instances on boot.
```

Command: `sudo systemctl status pce-mcp.service`

```
● pce-mcp.service - Model Context Protocol (MCP) server for Pextra
CloudEnvironment(R)
     Loaded: loaded (/lib/systemd/system/pce-mcp.service; enabled; preset:
enabled)
     Active: active (running) since Wed 2026-05-13 21:46:43 UTC; 1h 6min
ago
   Main PID: 642 (pce-mcp)
      Tasks: 9 (limit: 4636)
     Memory: 17.4M
        CPU: 70ms
     CGroup: /system.slice/pce-mcp.service
```

```
               └─642 /usr/bin/pce-mcp serve --tls-ca-cert=/etc/caddy/pce.crt
--base-url=https://localhost:5007 --disable-stdio --sse-addr= --http-
addr=127.0.0.1:7778

May 13 21:46:43 lab systemd[1]: Started pce-mcp.service - Model Context
Protocol (MCP) server for Pextra CloudEnvironment(R).
May 13 21:46:43 lab pce-mcp[642]: 2026/05/13 21:46:43 SSE server disabled
(empty address)
May 13 21:46:43 lab pce-mcp[642]: 2026/05/13 21:46:43 Serving HTTP at
127.0.0.1:7778
May 13 21:46:43 lab pce-mcp[642]: 2026/05/13 21:46:43 Stdio server disabled
```

Default Node Kernel: `BOOT_IMAGE=/boot/vmlinuz-6.1.0-43-amd64 root=UUID=2d61eb86-70bb-408a-bf4f-89eba7915b85 ro quiet`

No errors were found so only the head and tail of the kernel logs from Pextra are included.

```
kernel - Linux version 6.1.0-43-amd64 (debian-kernel@lists.debian.org)
(gcc-12 (Debian 12.2.0-14+deb12u1) 12.2.0, GNU ld (GNU Binutils for Debian)
2.40) #1 SMP PREEMPT_DYNAMIC Debian 6.1.162-1 (2026-02-08)
kernel - BIOS-provided physical RAM map:
kernel - BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
kernel - BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
kernel - BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
kernel - BIOS-e820: [mem 0x0000000000100000-0x00000000dffeffff] usable
kernel - BIOS-e820: [mem 0x00000000dfff0000-0x00000000dfffffff] ACPI data
kernel - BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
kernel - BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
kernel - BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
kernel - BIOS-e820: [mem 0x0000000100000000-0x000000011fffffff] usable
kernel - NX (Execute Disable) protection: active
kernel - SMBIOS 2.5 present.
kernel - DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox
12/01/2006
kernel - Hypervisor detected: KVM
kernel - kvm-clock: Using msrs 4b564d01 and 4b564d00
kernel - kvm-clock: using sched offset of 9229907771 cycles
kernel - clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles:
0x1cd42e4dffb, max_idle_ns: 881590591483 ns
kernel - tsc: Detected 3417.598 MHz processor
kernel - e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
kernel - e820: remove [mem 0x000a0000-0x000fffff] usable
kernel - last_pfn = 0x120000 max_arch_pfn = 0x400000000
kernel - Disabled
kernel - x86/PAT: MTRRs disabled, skipping PAT initialization too.
kernel - CPU MTRRs all blank - virtualized system.
kernel - x86/PAT: Configuration [0-7]: WB  WT  UC- UC  WB  WT  UC- UC
kernel - last_pfn = 0xdfff0 max_arch_pfn = 0x400000000
kernel - found SMP MP-table at [mem 0x0009fff0-0x0009ffff]
kernel - RAMDISK: [mem 0x328e1000-0x35467fff]
kernel - ACPI: Early table checksum verification disabled
kernel - ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX  )
```

```
kernel - ACPI: XSDT 0x00000000DFFF0030 00003C (v01 VBOX   VBOXXSDT 00000001
ASL  00000061)
kernel - ACPI: FACP 0x00000000DFFF00F0 0000F4 (v04 VBOX   VBOXFACP 00000001
ASL  00000061)
kernel - ACPI: DSDT 0x00000000DFFF0620 002353 (v02 VBOX   VBOXBIOS 00000002
INTL 20230628)
kernel - ACPI: FACS 0x00000000DFFF0200 000040
kernel - ACPI: FACS 0x00000000DFFF0200 000040
kernel - ACPI: APIC 0x00000000DFFF0240 00006C (v02 VBOX   VBOXAPIC 00000001
ASL  00000061)
kernel - ACPI: SSDT 0x00000000DFFF02B0 00036C (v01 VBOX   VBOXCPUT 00000002
INTL 20230628)
kernel - ACPI: Reserving FACP table memory at [mem 0xdfff00f0-0xdfff01e3]
kernel - ACPI: Reserving DSDT table memory at [mem 0xdfff0620-0xdfff2972]
kernel - ACPI: Reserving FACS table memory at [mem 0xdfff0200-0xdfff023f]
kernel - ACPI: Reserving FACS table memory at [mem 0xdfff0200-0xdfff023f]
kernel - ACPI: Reserving APIC table memory at [mem 0xdfff0240-0xdfff02ab]
kernel - ACPI: Reserving SSDT table memory at [mem 0xdfff02b0-0xdfff061b]
kernel - No NUMA configuration found
kernel - Faking a node at [mem 0x0000000000000000-0x000000011fffffff]
kernel - NODE_DATA(0) allocated [mem 0x11ffd1000-0x11fffbfff]
kernel - Zone ranges:
kernel -   DMA      [mem 0x0000000000001000-0x0000000000ffffff]
kernel -   DMA32    [mem 0x0000000001000000-0x00000000ffffffff]
kernel -   Normal   [mem 0x0000000100000000-0x000000011fffffff]
kernel -   Device   empty
kernel - Movable zone start for each node
kernel - Early memory node ranges
kernel -   node   0: [mem 0x0000000000001000-0x000000000009efff]
kernel -   node   0: [mem 0x0000000000100000-0x00000000dffeffff]
...
kernel - pci 0000:00:06.0: reg 0x10: [mem 0xf0804000-0xf0804fff]
kernel - pci 0000:00:07.0: [8086:7113] type 00 class 0x068000
kernel - pci 0000:00:07.0: quirk: [io  0x4000-0x403f] claimed by PIIX4 ACPI
kernel - pci 0000:00:07.0: quirk: [io  0x4100-0x410f] claimed by PIIX4 SMB
kernel - pci 0000:00:0b.0: [8086:265c] type 00 class 0x0c0320
kernel - pci 0000:00:0b.0: reg 0x10: [mem 0xf0805000-0xf0805fff]
kernel - pci 0000:00:0d.0: [8086:2829] type 00 class 0x010601
kernel - pci 0000:00:0d.0: reg 0x10: [io  0xd240-0xd247]
kernel - pci 0000:00:0d.0: reg 0x14: [io  0xd248-0xd24b]
kernel - pci 0000:00:0d.0: reg 0x18: [io  0xd250-0xd257]
kernel - pci 0000:00:0d.0: reg 0x1c: [io  0xd258-0xd25b]
kernel - pci 0000:00:0d.0: reg 0x20: [io  0xd260-0xd26f]
kernel - pci 0000:00:0d.0: reg 0x24: [mem 0xf0806000-0xf0807fff]
kernel - ACPI: PCI: Interrupt link LNKA configured for IRQ 11
kernel - ACPI: PCI: Interrupt link LNKB configured for IRQ 10
kernel - ACPI: PCI: Interrupt link LNKC configured for IRQ 9
kernel - ACPI: PCI: Interrupt link LNKD configured for IRQ 11
kernel - iommu: Default domain type: Translated
```

5. Observed Errors

Logs reviewed using the following commands:

Command: `journalctl -p err -xb`

```
May 13 21:46:42 lab systemd[1]: Invalid DMI field header.
May 13 21:46:42 lab kernel: platform regulatory.0: firmware: failed to load
regulatory.db (-2)
May 13 21:46:42 lab kernel: firmware_class: See
https://wiki.debian.org/Firmware for information about missing
May 13 21:46:42 lab kernel: platform regulatory.0: firmware: failed to load
regulatory.db (-2)
May 13 21:46:42 lab kernel: [drm:vmw_host_printf [vmwgfx]] *ERROR* Failed
to send host log message.
May 13 21:46:44 lab libvirtd[639]: Unable to open /dev/kvm: No such file or
directory
May 13 21:46:53 lab libvirtd[639]: Unable to open /dev/kvm: No such file or
directory
May 13 21:46:53 lab libvirtd[639]: Unable to open /dev/kvm: No such file or
directory
```

Command: `dmesg -l err`

```
[    1.944541] systemd[1]: Invalid DMI field header.
[    2.346097] platform regulatory.0: firmware: failed to load
regulatory.db (-2)
[    2.346568] firmware_class: See https://wiki.debian.org/Firmware for
information about missing firmware
[    2.347133] platform regulatory.0: firmware: failed to load
regulatory.db (-2)
[    2.477587] [drm:vmw_host_printf [vmwgfx]] *ERROR* Failed to send host
log message.
```

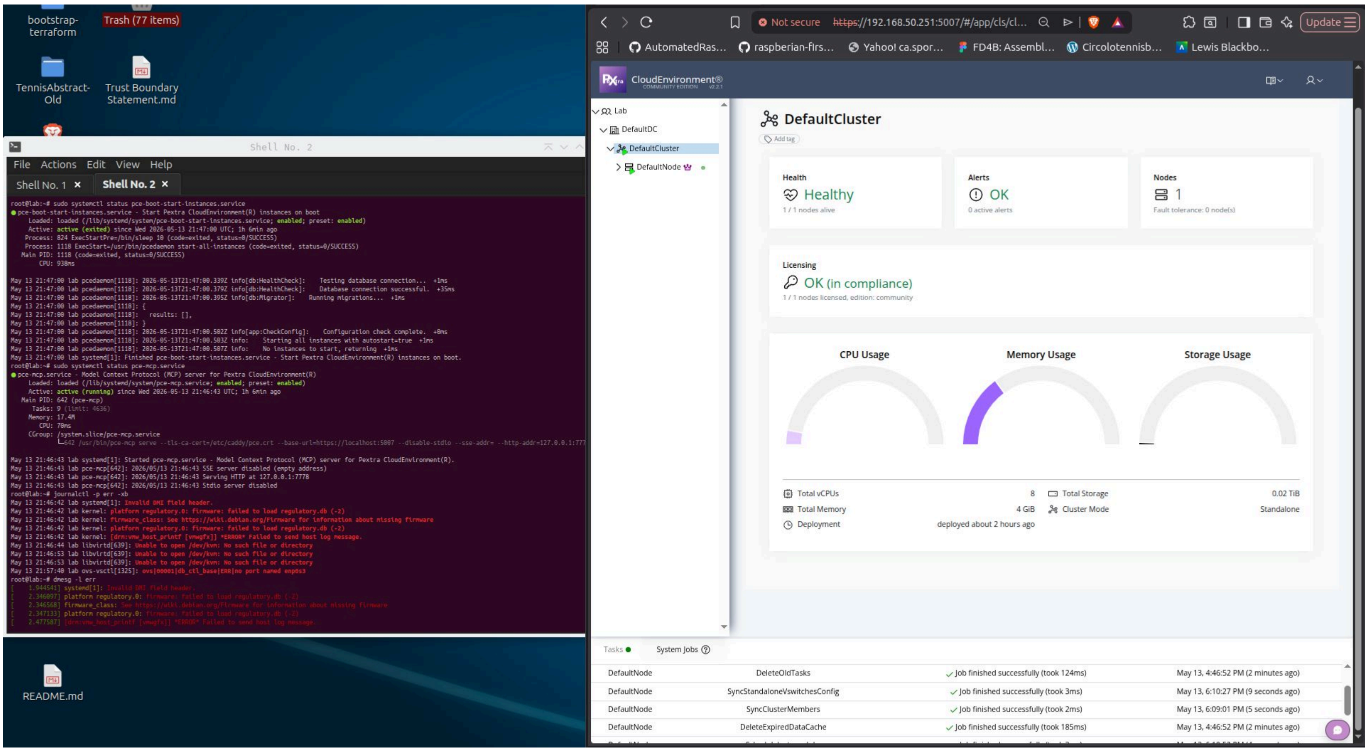
None of these errors are critical for the most part and the only one that may need investigation is the `Unable to open /dev/kvm: No such file or directory` log message to determine if any process or service is being blocked. Since I tested in a VM, nested virtualization is not available inside the VM.

`vmwgfx Failed to send host log message` is related to virtual GPU subsystem and is non-critical in headless server use, but I'm not sure if that matters and it depends on how on-prem access is determined.

6. Conclusion

The Pextra OS image is compatible with the KVM virtualization environment for standard deployment and management usage. No critical system failures or application-level issues were observed. The system is stable under current test conditions.

Nested virtualization is not supported in this configuration, but this does not impact core Pextra dashboard functionality.



7. Recommendations & Next Steps

1. Ready for testing on-prem once a VLAN is provided for network setup

2. Suitable for use as a management/control-plane VM in a lab environment after confirmation of no errors on on-prem hardware as well

3. No blocking compatibility issues identified during proof of concept test

4. Required to get the networking info before bootstrapping the control plane or management interface for production